The 7th Annual

Parallel Applications Workshop
Alternatives to MPI+X

Sunday, November 17, 2024
9:00 am - 5:30 pm (EST)

Hybrid Event

Held in conjunction with SC24
The International Conference for High Performance Computing, Networking, Storage, and Analysis

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

Organization

- Workshop Chair
  - Karla V. Morris Wright - Sandia National Laboratories
- Organizing Committee
  - Engin Kayraklioglu - Hewlett Packard Enterprise
  - Kenjiro Taura - University of Tokyo
  - Irene Moulitsas - Cranfield University
  - Elliott Slaughter - SLAC National Accelerator Laboratory
- Program Committee Chairs
  - Bill Long - Hewlett Packard Enterprise
  - Daniele Lezzi - Barcelona Supercomputing Center
- Advisory Committee
  - Bradford L. Chamberlain - Hewlett Packard Enterprise
  - Damian W. I. Rouson - Lawrence Berkeley National Laboratory

SC24 Atlanta, GA | hpc creates.

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

Program Committee

Chair: Daniele Lezzi	- Barcelona Supercomputing Center
Chair: Bill Long	- Hewlett Packard Enterprise

Marjan Asgari	- National Resources Canada
Scott Baden	- University of California, San Diego
Dan Bonachea	- Lawrence Berkeley National Laboratory
Jan Ciesko	- Sandia National Laboratory
Nelson Dias	- Federal University of Paraná
Mario Di Renzo	- University of Salento
David Eberius	- Intel
Engin Kayraklioglu	- Hewlett Packard Enterprise
Francesc Lordan	- Barcelona Supercomputing Center
Henry Monge Camacho	- Oak Ridge National Laboratory
Karla Morris	- Sandia National Laboratories
Irene Moulitsas	- Cranfield University
Catherine Olschanowsky	- Advanced Micro Devices, Inc
Tom Quinn	- University of Washington
Michel Schanen	- Argonne National Laboratory
Michael Schlottke-Lakemper	- University of Augsburg
Elliott Slaughter	- SLAC National Accelerator Laboratory
Kenjiro Taura	- University of Tokyo
Thiago Teixeira	- Intel
Jana Thayer	- SLAC National Accelerator Laboratory
Miwako Tsuji	- RIKEN

SC24
Atlanta, GA | hpc creates.

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

Artifact Evaluation Committee

- **Chair:** Irene Moulitsas - Cranfield University
- **Chair:** Elliott Slaughter - SLAC National Accelerator Laboratory

- Oliver Alvarado Rodriguez - New Jersey Institute of Technology
- Desmond Bisandu - Cranfield University
- Yakup Budanaz - ETH Zurich
- Fabio Durastante - University of Pisa
- Guillaume Helbecque - University of Lille
- Mert Hidayetoglu - Stanford University
- Boyu Neil Kuang - Cranfield University
- Seema Mirchandaney - Stanford University
- Soren Rasmussen - National Center for Atmospheric Research
- Kate Rasmuseen - Lawrence Berkeley National Laboratory
- Anjiang Wei - Stanford University

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

Program

9:00 - Welcome and Introduction
9:02 - **Session 1**
10:00 - Break (30 minutes)
10:30 - **Session 2**
12:30 - Lunch (90 minutes)

2:00 - **Session 3: Distinguished Talk**
Éric Laurendeau - Polytechnique Montréal
A Case Study for Using Chapel
within the Global Aerospace Industry

3:30 - Break (30 minutes)
3:30 - **Session 4**
3:45 - **Panel Discussion**
5:30 - PAW-ATM Concludes

SC24 Atlanta, GA | hpc creates.

PAW-ATM

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

SC24
Atlanta, GA | hpc creates.

Session 1

9:00 - 10:00

- **9:02: Survey of Technologies for Developers of Parallel Applications**

  Wonchan Lee – NVIDIA
  Damian Rouson – Lawrence Berkeley National Laboratory

- **9:45: User Experience: *Resource Adaptivity at Task-Level***

  Jonas Posner

PAW-ATM

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

SC24 Atlanta, GA | hpc creates.

Session 2

10:30 - 12:30

- **10:30*: Speeding-Up LULESH on HPX: Useful Tricks and Lessons Learned using a Many-Task-Based Approach***

  Torben Kalkhof, and Andreas Koch

- **10:50: *Lamellar: A Rust-based Asynchronous Tasking and PGAS Runtime for High Performance Computing***

  Ryan D. Friese, Roberto Gioiosa, Joseph Cottam, Erdal Mutlu, Gregory Henselman-Petrusek, Polykarpos Thomadakis, Mark Raugas

- **11:10: *Applying a Task-Based Approach to Distributed Machine Learning Workflows***

  Fernando Vazquez-Novoa, Daniele Lezzi, Francesc Lordan, Fatemeh Baghdadi, and Davide Cirillo

- **11:30: *Accelerating Multi-GPU Embedding Retrieval with PGAS style Communication for Deep Learning Recommendation Systems***

  Yuxin Chen, Aydin Buluc, Katherine Yelick, and John Owens

- **11:50: *Mitigating Synchronization Bottlenecks in High-Performance Actor-Model-Based Software***

  Kyle Klenk, Mohammad Mahdi Moayeri, Junwei Guo, Martyn P. Clark, and Raymond J. Spiteri

- **12:10: *Intel SHMEM: GPU-initiated OpenSHMEM using SYCL***

  Alex Brooks, Philip Marshall, David Ozog, Md. Wasi-ur- Rahman, Lawrence Stewart, and Rithwik Tom

Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday

SC24
Atlanta, GA
hpc creates.

Session 3

2:00 - 3:00

2:00: Distinguished Talk

A Case Study for Using Chapel within the Global Aerospace Industry

Éric Laurendeau, Polytechnique Montréal

- **2:45: User Experience: Exploring Suffix Array Algorithms in Chapel**

  Michael P. Ferguson, Bonnie Hurwitz, and Shreyas Khandekar

Parallel Applications Workshop Alternatives to MPI+X

November 17th, 2024
Sunday

SC24 Atlanta, GA | hpc creates.

Session 4

3:30 - 5:30

- **3:30: User Experience: *Just Write Fortran: Experiences with a Language-Based Alternative to MPI+X***

Baboucarr Dibba, Katherine Rasmussen, Brad Richardson, Damian Rouson, David Torres, Yunhao Zhang, Ethan Gutmann, Kareem Ergawy and Michael Klemm

- **3:45: Panel: Alternative programming models for applications at scale**

Panel Chair: Christine Sweeney – Los Alamos National Laboratory

- Jan Ciesko – Sandia National Laboratories
- Nils Deppe – Cornell University
- Jason DeVinney – Center for Computing Sciences
- Eric Laurendeau – Polytechnique Montreal
- Julian Samaroo – Massachusetts Institute of Technology

PAW-ATM Journey

PAW-ATM

SC24

Atlanta, GA | hpc creates.

PAW ATM

SC23

Denver, CO | I am hpc.

PAW ATM

SC22

PAW ATM

SC21

St. Louis, MO | science & beyond.

PAW-ATM

SC20

PAW-ATM

SC19

Denver, CO | hpc is now.

PAW-ATM

SC18

Dallas, TX | hpc inspires.

PAW

SC17

Denver, CO | hpc connects.

PAW

SC16

Salt Lake City, Utah | hpc matters.